

JEE-Main-31-01-2024 (Memory Based)

[MORNING SHIFT]

Physics

Question: Find the energy released in kilowatt-hour if 4 gram of mass converts to energy.

Given speed of light is

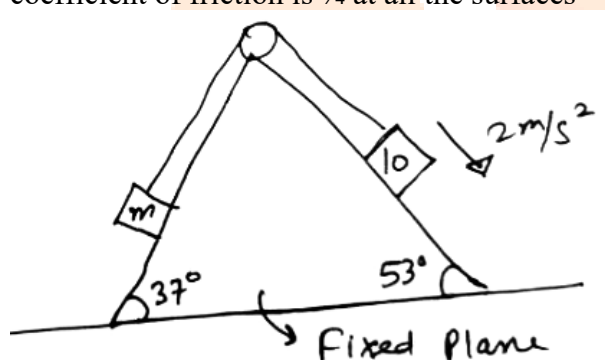
$$C = 3 \times 10^8 \text{ m/s}$$

Options:

- (a) 10^6 Kwh
- (b) 10^7 Kwh
- (c) 10^8 Kwh
- (d) 10^9 Kwh

Answer: (c)

Question: Find the value of m if the 10 kg block accelerates at 2 m/s^2 down the plane when coefficient of friction is $\frac{1}{4}$ at all the surfaces

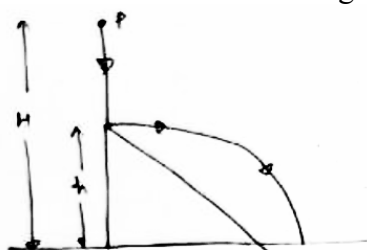


Options:

- (a) 4.5 kg
- (b) 3.5 kg
- (c) 2.5 kg
- (d) 1.5 kg

Answer: (a)

Question: A particle 'P' is released from a fixed height 'H' from the ground if falls on an incline at its top point. After striking the plane its velocity becomes horizontal. Find the ratio H/h so that total time of flight becomes maximum

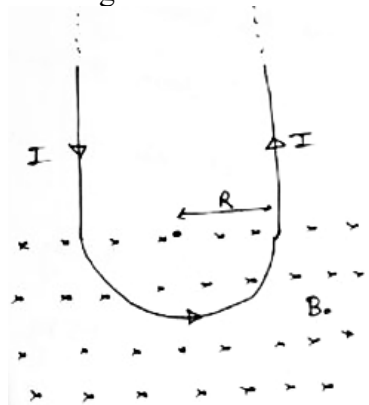


Options:

- (a) 4
- (b) 6
- (c) 2
- (d) 8

Answer: (c)

Question: A uniform magnetic field B_0 exists going inside the screen as shown. The two infinite parallel wires are joined to the semicircular wire of radius R as shown, and carry current I . Find the force on the wire if only the semicircular wire is present in the magnetic field region.



Options:

- (a) $2 IRB_0$
- (b) $I RB_0$
- (c) $4 I RB_0$
- (d) 0

Answer: (a)

Question: A capacitor connected with a battery has a charge Q . The charge increases to $2.5Q$ if a dielectric material with dielectric constant ' k ' is inserted. The value of ' k ' is

Options:

- (a) 0.4
- (b) 2.5
- (c) 1
- (d) 5

Answer: (b)

Question: In a region of space, the peak electric field due to electromagnetic wave is 50 N/C . Find the average energy density in this region.

Options:

- (a) $1.1 \times 10^{-8} \text{ J/m}^3$
- (b) $2.1 \times 10^{-8} \text{ J/m}^3$
- (c) $5.5 \times 10^{-9} \text{ J/m}^3$
- (d) $2.2 \times 10^{-8} \text{ J/m}^3$

Answer: (a)

Question: Two charges ' q ' and $3q$ are separated by a distance ' r ' if ' x ' is the distance from ' q ' where the electric field is zero, find x

Options:

- (a) $0.37r$

- (b) $0.73r$
 (c) $1.7r$
 (d) $1.4r$

Answer: (a)

Question: Light from two sources of intensities in ratio 1:9 and phase difference 60 degrees meet on the screen. Find the ratio of net intensity when they are coherent versus when they are not

Options:

- (a) $8/5$
 (b) $5/8$
 (c) $13/10$
 (d) $10/13$

Answer: (c)

Question: Bulk modulus of rubber is $9 \times 10^8 \text{ N/m}^2$. To what depth a rubber ball is taken so that its volume decreases by 0.02 %. Density of water 10^3 kg/m^3 , $g = 10 \text{ m/s}^2$

Options:

- (a) 5 m
 (b) 18 m
 (c) 180 m
 (d) 100

Answer: (b)

Question: Two equal resistances with temperature coefficients of resistances α_1 and α_2 respectively are first joined in series and then joined in parallel. The equivalent temperature coefficients in the two cases will be

Options:

- (a) $\alpha_1 + \alpha_2, \frac{\alpha_1 + \alpha_2}{2}$
 (b) $\frac{\alpha_1 + \alpha_2}{2}, \frac{\alpha_1 + \alpha_2}{2}$
 (c) $\frac{\alpha_1 + \alpha_2}{2}, \alpha_1 + \alpha_2$
 (d) $\alpha_1 + \alpha_2, \alpha_1 + \alpha_2$

Answer: (b)

Question: If $t = \alpha x^2 + \beta x$ & v sphere velocity. Find Acceleration of particle

Options:

Answer: $(2\alpha v^3)$

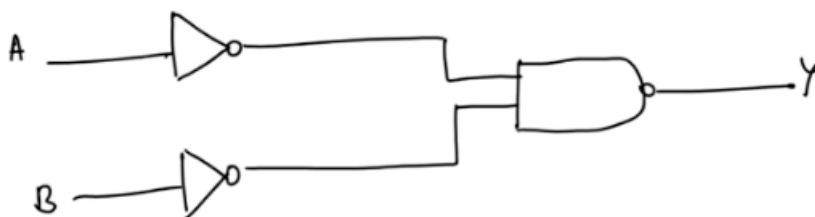
Question: If the error in the length and cross sectional diameter of a wire is both 1%, then the error in the resistance will be:

Options:

- (a) 2%
 (b) 3%
 (c) 4%
 (d) 5%

Answer: (b)

Question: What is the below logic gate circuit equivalent to



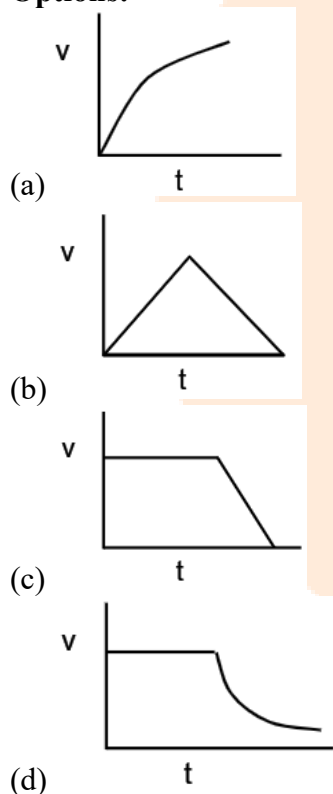
Options:

- (a) AND gate
- (b) OR gate
- (c) NAND gate
- (d) NOR gate

Answer: (b)

Question: A spherical ball is released from rest in a long cylinder filled with glycerine. The graph of its velocity vs time is

Options:



Answer: (a)

Question: A Prism has a refractive index of $\mu = \cot \frac{A}{2}$. Find the minimum deviation if it is kept in air. A is angle of Prism

Prism

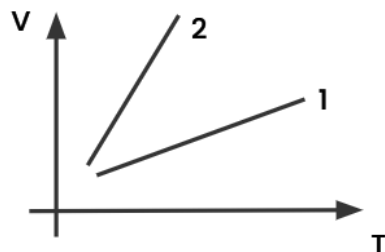
Options:

- (a) $\frac{\pi}{2} - \frac{A}{2}$

- (b) $\pi - 2A$
 (c) $\pi - A$
 (d) $\frac{\pi}{2} - A$

Answer: (b)

Question: Given two isobaric process compare the pressures ?



Options:

- (a) $P_1 < P_2$
 (b) $P_1 > P_2$
 (c) $P_1 = P_2$
 (d) None of the above

Answer: (b)

Question: Four particles each of mass 'm' are placed at the four vertices of a square of side 'a'. Find the net force on any one of the particle ?

Options:

Answer: $\frac{G.m^2}{2a^2}(2\sqrt{2} + 1)$

Question: If the stopping potential for light of wavelength λ is 8 Volts then find the threshold wavelength if the stopping potential becomes 2 Volts for a wavelength of 3λ ?

Options:

- (a) λ
 (b) 2λ
 (c) 3λ
 (d) 9λ

Answer: (d)

Question: An artillery of mass M_1 carries a shell of M_2 mass. Initially both are at rest. If artillery fires the shell on smooth ground in the horizontal direction then the ratio of KE of artillery & shell will be

Options:

- (a) M_1/M_2
 (b) M_2/M_1
 (c) M_1+M_2/M_2
 (d) M_1+M_2/M_1

Answer: (b)

Question: A block is performing SHM of amplitude A. When it is at $2A/3$ from mean position, its velocity is tripled. Find the new amplitude of motion. $A' = \frac{A}{3}$

Options:

- (a) 7
- (b) 6
- (c) 5
- (d) 3

Answer: (a)



JEE-Main-31-01-2024 (Memory Based)
[MORNING SHIFT]

Chemistry

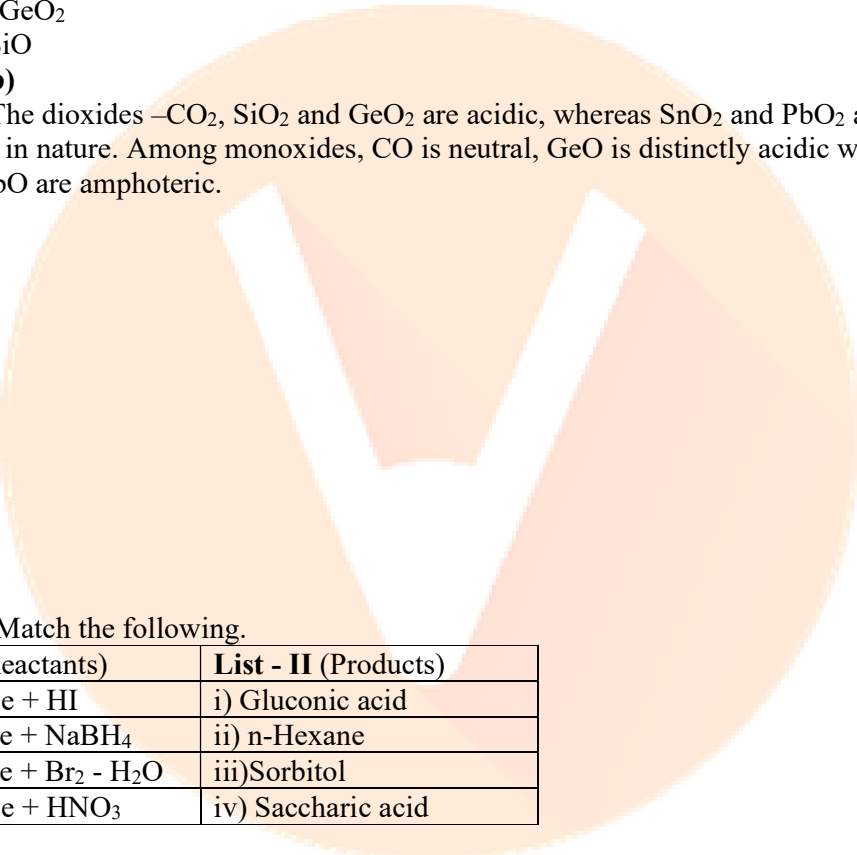
Question: Which of the following is amphoteric?

Options:

- (a) GeO & GeO_2
- (b) SnO_2 & PbO_2
- (c) SiO_2 & GeO_2
- (d) CO & SiO

Answer: (b)

Solution: The dioxides CO_2 , SiO_2 and GeO_2 are acidic, whereas SnO_2 and PbO_2 are amphoteric in nature. Among monoxides, CO is neutral, GeO is distinctly acidic whereas SnO and PbO are amphoteric.



6 C
14 Si
32 Ge
50 Sn
82 Pb

Question: Match the following.

List - I (Reactants)	List - II (Products)
A) Glucose + HI	i) Gluconic acid
B) Glucose + NaBH_4	ii) n-Hexane
C) Glucose + $\text{Br}_2 - \text{H}_2\text{O}$	iii) Sorbitol
D) Glucose + HNO_3	iv) Saccharic acid

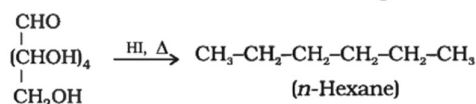
Options:

- (a) A - iii, B - ii; C - i; D - iv
- (b) A - ii, B - iii; C - i; D - iv
- (c) A - i, B - iii; C - ii; D - iv
- (d) A - ii, B - iii; C - iv; D - i

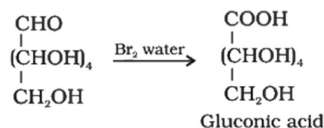
Answer: (b)

Solution:

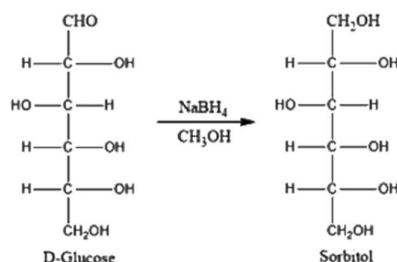
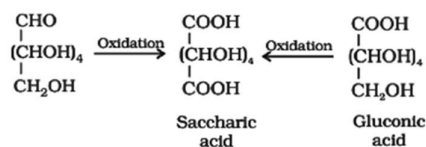
On prolonged heating with HI, it forms n-hexane, suggesting that all the six carbon atoms are linked in a straight chain.



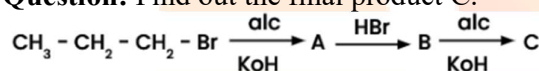
Glucose gets oxidised to six carbon carboxylic acid (gluconic acid) on reaction with a mild oxidising agent like bromine water. This indicates that the carbonyl group is present as an aldehydic group.



On oxidation with nitric acid, glucose as well as gluconic acid both yield a dicarboxylic acid, saccharic acid. This indicates the presence of a primary alcoholic (-OH) group in glucose.



Question: Find out the final product C.

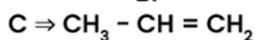
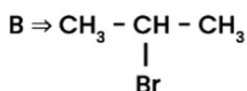
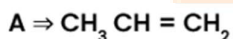


Options:

- (a) Propan - 1 - ol
- (b) Propan - 2 - ol
- (c) Propene
- (d) Propane

Answer: (c)

Solution:



Question: Which Compound is white in colour in aqueous medium?

Options:

- (a) ZnSO_4
- (b) CuSO_4
- (c) FeSO_4
- (d) FeCl_3

Answer: (a)

Solution: Zn salts are colourless due to completely filled d orbital.

Question: On which factor electrical conductivity of electrolytic cell does not depend

Options:

- (a) Concentration of electrolyte

- (b) Amount of electrolyte added
- (c) Temperature
- (d) Nature of electrode

Answer: (d)

Solution: It depends on the nature of the electrolyte and concentration of the electrolyte.

Question: Decreasing order of electron gain enthalpy of the following elements (magnitude only)

Sulphur - A, Bromine - B, Fluorine - C, Argon - D

Options:

- (a) $A > B > C > D$
- (b) $D > C > B > A$
- (c) $C > B > A > D$
- (d) $A > B > D > C$

Answer: (c)

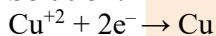
Question: If one faraday of electricity is used in the discharging of Cu^{2+} , then find the mass (in g) of Cu deposited

Options:

- (a) 31.75 g
- (b) 45.9 g
- (c) 65.3 g
- (d) 27.5 g

Answer: (a)

Solution:



Two moles of electrons are required to deposit one mole Cu. Therefore, the weight of copper that will get deposited on passing 2 faradays of electricity is 63.5g.

$$1\text{F} = 63.5/2 \text{ g}$$

$$31.75\text{g}$$

Question: Statement I: Dichromates are generally made from cromates

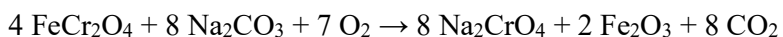
Statement II: Manganate ions are diamagnetic

Options:

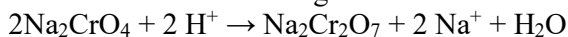
- (a) Both statement I and statement II are false
- (b) Statement I is true but statement II is false
- (c) Statement I is false but statement II is true
- (d) Both statement I and statement II are true

Answer: (b)

Solution: Potassium dichromate is a very important chemical used in leather industry and as an oxidant for preparation of many azo compounds. Dichromates are generally prepared from chromate, which in turn are obtained by the fusion of chromite ore (FeCr_2O_4) with sodium or potassium carbonate in free access of air. The reaction with sodium carbonate occurs as follows:



The yellow solution of sodium chromate is filtered and acidified with sulphuric acid to give a solution from which orange sodium dichromate, $\text{Na}_2\text{Cr}_2\text{O}_7 \cdot 2\text{H}_2\text{O}$ can be crystallised.



Question: Which have highest electron gain enthalpy?

Options:

- (a) F
- (b) Cl
- (c) Br
- (d) I

Answer: (b)

Solution: Electron-electron repulsion is very high in fluorine which resists the addition of an electron. Thus, electron gain enthalpy follows the order: $\text{Cl} > \text{F} > \text{Br} > \text{I}$.

Question: Which of the following give positive deviation from Raoult's Law ?

Options:

- (a) Ethanol + acetone
- (b) Benzene + toluene
- (c) Acetone + chloroform
- (d) Chloroethane + bromoethane

Answer: (b)

Question: Assertion: Noble gas have very high boiling point.

Reason: Noble gas have weak dispersion forces

Options:

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true but R is not the correct explanation of A.
- (c) A is true but R is false.
- (d) A is false but R is true.

Answer: (d)

Solution: Noble gases have weak Vander waals forces present in them both in the liquid and solid state. and have very low melting and boiling point due to this reason.

Question: Statement-I: Pka value of Phenol and ethanol is 10.0 and 15.9 respectively.

Statement-II: Ethanol is more Acidic than phenol.

Options:

- (a) Both statement I and statement II are false
- (b) Statement I is true but statement II is false
- (c) Statement I is false but statement II is true
- (d) Both statement I and statement II are true

Answer: (b)

Solution: Greater the pK_a value. Weaker is the acid Hence, phenol is more acidic than ethanol.

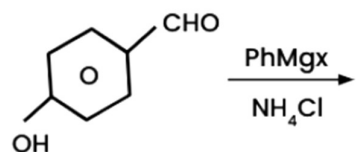
Question: Which of the following does not give colour with conc. H_2SO_4 ?

Options:

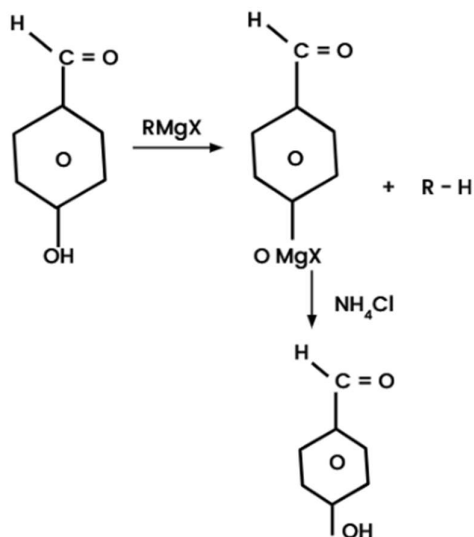
- (a) NaBr
- (b) CaF_2
- (c) NaNO_3
- (d) I⁻

Answer: (a)

Question:



Solution:



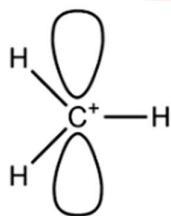
Question: Which of the following have six electron in carbon?

Options:

- (a) Carbocation
- (b) Carbanion
- (c) Carbon free radical
- (d) None of the above

Answer: (a)

Solution:



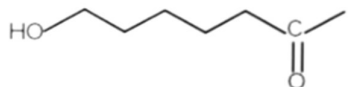
Question: Adsorption method is used in.

Options:

- (a) Chromatography
- (b) Extracational method
- (c) Distillation method
- (d) Sublimation

Answer: (a)

Question: Correct IUPAC name of

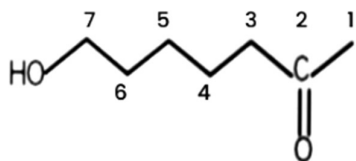


Options:

- (a) 7 - Hydroxyheptan - 2 - one
- (b) 6 - Hydroxyheptan - 2 - one
- (c) 2 - Oxoheptan - 7 - ol
- (d) 1 - Hydrogen - 6 - oxoheptane

Answer: (a)

Solution:



Question: White colour compound is

Options:

- (a) Ammonium molybdate
- (b) Ammonium sulphide
- (c) Lead sulphate
- (d) Lead iodide

Answer: (c)

Question: Statement I: Alcohols can act as nucleophile as well as electrophile

Statement II: Alcohols react with metals to form alkoxide and liberate H_2 .

Options:

- (a) Both statement I and statement II are false
- (b) Statement I is true but statement II is false
- (c) Statement I is false but statement II is true
- (d) Both statement I and statement II are true

Answer: (c)

Solution:

Statement 2 true

Statement 1 alcohol act as nucleophile and protonated alcohol act as electrophile

Question: How many of the following compounds have sp^3 hybridized central atom?

H_2O , NH_3 , SiO_2 , SO_2 , CO and BF_3

Answer: 3

Solution:



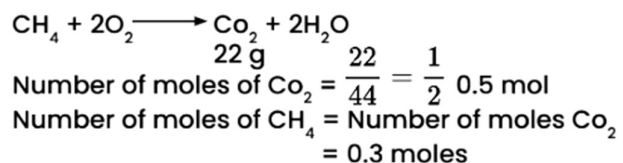
sp^3
hybrid
orbitals

Question: Moles of CH_4 required for formation of 22 g of CO_2 is $m \times 10^{-2}$

The value of m is :

Answer: 50

Solution:



Question: The total number of different alkanes formed when the following mixture is subjected to electrolysis :

CH_3COONa (aq) and $\text{C}_2\text{H}_5\text{COONa}$ (aq) is _____
 (do not consider disproportionation reaction)

Answer: 3

Solution: Ethane, Butane & propane

Question: Which of the following are generally used in batteries?

Zn, Cd, Hg, Mn, Fe

Answer: 4

Question: Number of Geometrical Isomers of $[\text{Pt}(\text{en})_2 \text{Cl}_2]$

Answer: 2

JEE-Main-31-01-2024 (Memory Based)

[MORNING SHIFT]

Mathematics

Question: Solve differential equation:

$$\frac{dx}{dy} = x(\ln x - \ln y + 1)$$

Answer:

Solution:

$$\frac{dx}{dy} = \frac{x}{y} \left(\ln \frac{x}{y} + 1 \right)$$

Put $x = vy$

$$\Rightarrow v + y \frac{dv}{dy} = v(\ln v + 1)$$

$$\Rightarrow \int \frac{dv}{v \ln v} = \int \frac{dy}{y}$$

$$\Rightarrow \ln |\ln v| = \ln |y| + \ln c$$

$$\Rightarrow \left| \ln \frac{x}{y} \right| = c |y|$$

Question: $\lim_{x \rightarrow 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2} = ?$

Options:

- (a) Does not exist
- (b) 2
- (c) 1
- (d) -1

Answer: (b)

Solution:

$$\lim_{x \rightarrow 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{|\sin x|^2} \times \frac{\sin^2 x}{x^2}$$

$$= \lim_{t \rightarrow 0} \frac{e^{2t} - 2t - 1}{t^2} \quad t = |\sin x|$$

$$= \lim_{t \rightarrow 0} \frac{2e^t - 2}{2t} = \lim_{t \rightarrow 0} \frac{2(e^{2t} - 1)}{2t} = 2$$

Question: $|\vec{a}| = 1, |\vec{b}| = 4, \vec{a} \cdot \vec{b} = 2, \vec{c} = (2\vec{a} \times \vec{b} - 3\vec{b})$. If angle between $\vec{c}$ and $\vec{b}$ is α then find $192 \sin^2 \alpha$.

Answer: 48.00

Solution:

$$|\vec{a}| = 1, |\vec{b}| = 4, \vec{a} \cdot \vec{b} = 2 \rightarrow \cos \theta = \frac{1}{2} \rightarrow \theta = \frac{\pi}{3}$$

$$\vec{c} = 2\vec{a} \times \vec{b} - 3\vec{b}$$

$$|\vec{c}|^2 = 4|\vec{a} \times \vec{b}|^2 + 9|\vec{b}|^2$$

$$|\vec{c}|^2 = 4|\vec{a} \times \vec{b}|^2 + 9|\vec{b}|^2$$

$$= 4|\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta + 9|\vec{b}|^2$$

$$= 4 \times 1 \times 16 \times \frac{3}{4} + 9 \times 16$$

$$= 48 + 144 = 192$$

$$\vec{b} \cdot \vec{c} = 0 - 3|\vec{b}|^2$$

$$4 \times 8 \sqrt{3} \cos \alpha = -3 \times 16$$

$$\cos \alpha = \frac{-\sqrt{3}}{2}$$

$$192 \sin^2 \alpha = 48$$

Question: Let 'S' be the set of positive integral values of a for which $\frac{ax^2 + 2(a+1)x + 9a + 4}{x^2 + 8x + 32} < 0, \forall x \in R$. Then, the number of elements in 'S' is

Answer:

Solution:

$$\frac{a^2 + 2(a+1)x + 4a}{x^2 + 8x + 32} < 0 \forall x$$

$$D^r > 0 \forall x$$

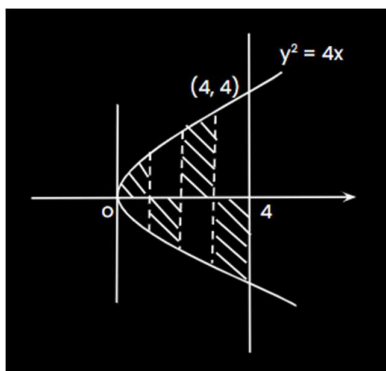
N^r can be -ve for all x only if

a is -ve so no +ve integral value of a .

Question: There was on area $\left\{ y^2 < 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} < 0 \right\}$.

Answer: $\frac{32}{3}$

Solution:



$$y > 0$$

$$x(x-1)(x-2)(x-3)(x-4) > 0$$

$$y < 0$$

$$x(x-1)(x-2)(x-3)(x-4) < 0$$

$$\text{Area} = \frac{2}{3} \times 16 = \frac{32}{3}$$

Question: If $f(x) = \begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-2 \end{vmatrix}$ for all $x \in \mathbb{R}$, then $2f(0) + f'(0)$ is equal to

Answer: 42.00

Solution:

$$f(0) = \begin{vmatrix} 0 & 1 & 1 \\ 2 & 0 & 6 \\ 0 & 4 & -2 \end{vmatrix} = 8 + 4 = 12$$

$$f'(0) = \begin{vmatrix} 0 & 0 & 3 \\ 2 & 0 & 6 \\ 0 & 4 & -2 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 4 & -2 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 1 \\ 2 & 0 & 6 \\ -1 & 0 & 0 \end{vmatrix}$$

$$= 24 + 0 + (-6) = 18$$

$$2f(0) + f'(0) = 42$$

Question: If $f(x) = \frac{4x+3}{6x-4}, x \neq \frac{2}{3}$. Find $(f \circ f)(x) = g(x)$, where $g: \mathbb{R} - \left\{\frac{2}{3}\right\} \rightarrow \mathbb{R} - \left\{\frac{2}{3}\right\}$

then $(g(g(g(4))))$ is equal to

Answer: 4.00

Solution:

$$f(x) = \frac{4x+3}{6x-4}$$

$$g(x) = f(f(x)) = \frac{4\left(\frac{4x+3}{6x-4}\right) + 3}{6\left(\frac{4x+3}{6x-4}\right) - 4}$$

$$= \frac{16x+12+18x-12}{24x+18-24x+16} = \frac{34x}{34} = x$$

$$g(g(g(4))) = 4$$

Question: $\sum_{r=1}^{10} \frac{r}{1-3r^2+r^4} = S$. Find S.

Answer: $\frac{-55}{109}$

Solution:

$$\sum_{r=1}^{10} \frac{r}{1-3r^2+r^4} = \sum_{r=1}^{10} \frac{r}{(r^2-1)^2 - r^2}$$

$$= \sum_{r=1}^{10} \frac{r}{(r^2+r-1)(r^2-r-1)}$$

$$= \sum_{r=1}^{10} \frac{1}{2} \left[\frac{1}{r^2-r-1} - \frac{1}{r^2+r-1} \right]$$

$$= \frac{1}{2} \left[\left(\frac{1}{-1} - \frac{1}{1} \right) + \left(\frac{1}{1} - \frac{1}{5} \right) + \dots + \left(-\frac{1}{109} \right) \right]$$

$$= \frac{1}{2} \left[-1 - \frac{1}{109} \right] = \frac{-110}{2 \times 109} = \frac{-55}{109}$$

Question: If the system of linear equation $x - 2y + z = -4$; $2x + \alpha y + 3z = 5$ and $3x - y + \beta z = 3$ has infinitely many solutions then $12\alpha + 13\beta$ is equal

Answer: 58.00

Solution:

$$x - 2y + z = 4$$

$$2x + \alpha y + 3z = 5$$

$$3x - y + \beta z = 3$$

$$\begin{vmatrix} 1 & -2 & 1 \\ 2 & \alpha & 3 \\ 3 & -1 & \beta \end{vmatrix} = \alpha - 18 - 2 - 3\alpha + 3 - 4\beta = 0$$

$$\alpha\beta - 3\alpha + 4\beta - 17 = 0$$

$$(\alpha + 4)(\beta - 3) = 5$$

$$\text{Also, } D_2 = \begin{vmatrix} 1 & -4 & 1 \\ 2 & 5 & 3 \\ 3 & 3 & \beta \end{vmatrix} = 0$$

$$\Rightarrow s\beta - 36 + 6 - 15 - 9 + 8\beta = 0$$

$$13\beta - 54 = 0$$

$$D_3 = \begin{vmatrix} 1 & -2 & -4 \\ 2 & \alpha & 5 \\ 3 & -1 & 3 \end{vmatrix} = 0$$

$$\rightarrow 3\alpha - 30 + 8 + 12\alpha + 5 + 12 = 0$$

$$15\alpha - 5 = 0 \Rightarrow \alpha = \frac{1}{3}$$

$$12\alpha + 13\beta = 4 + 54 = 58$$

$$\Delta = \left(\frac{1}{3} + 4\right) \left(\frac{54}{13} - 3\right)$$

$$= \frac{13}{3} \times \frac{15}{13} = 5$$

Also,

$$D_1 = \begin{vmatrix} -4 & -2 & 1 \\ 5 & \alpha & 3 \\ 3 & -1 & \beta \end{vmatrix}$$

$$= -4\alpha\beta - 18 - 5 - 3\alpha - 12 + 10\beta$$

$$= -4\alpha\beta - 3\alpha + 10\beta - 35$$

$$= -\frac{4\beta}{3} - 1 + 10\beta - 35$$

$$= \frac{26\beta}{3} - 36 = 0$$

Question: $f(x) = \begin{cases} g(x), & x \leq 0 \\ \left(\frac{x+1}{x+2}\right)^{\frac{1}{x}}, & x > 0 \end{cases}$, where $g(x)$ is linear function and $f(x)$ is

continuous at $x = 0$. Also $f'(1) = g(-1)$, $g(0) = f(0)$ then find the value of $g(3)$?

If ans is $\frac{a}{b} \ln\left(\frac{\alpha}{\beta} e^{\frac{1}{3}}\right)$ find $\alpha + \beta + a + b$

Answer: 15.00

Solution:

$$f(x) = \begin{cases} g(x), & x \leq 0 \\ \left(\frac{x+1}{x+2}\right)^{\frac{1}{x}}, & x > 0 \end{cases}$$

$$f(0^+) = 0; f(0^-) = g(0^-) = 0$$

$$\Rightarrow g(x) = mx$$

$$\begin{aligned} f'(1) &= \frac{d}{dx} e^{\frac{1}{x} [\ln(x+1) - \ln(x+2)]} \\ &= \frac{2}{3} \times \left[1 \cdot \left(\frac{1}{2} - \frac{1}{3} \right) - 1 \left(\ln \frac{2}{3} \right) \right] \\ &= \frac{2}{3} \left[\frac{1}{6} - \ln \frac{2}{3} \right] = \frac{1}{9} - \frac{2}{3} \ln \frac{2}{3} \\ &= g(-1) = -m \end{aligned}$$

$$g(3) = 3m = 2 \ln \frac{2}{3} - \frac{1}{3}$$

$$= 2 \left[\ln \frac{2}{3} - \ln e^{\frac{1}{3}} \right]$$

$$= 2 \left[\ln \frac{2}{2e^{\frac{1}{3}}} \right]$$

$$= \ln \frac{4}{9e^{\frac{1}{3}}}$$

$$a = b = 1, \alpha = 4, \beta = 9$$

$$\alpha + \beta + a + b = 15$$

Question: 3 rotten apples are mixed with 15 normal apples. Let the random variable be defined as number of rotten apples on picking 3 apples with replacement. Find variance of x

Answer: $\frac{5}{12}$

Solution:

3 rotten apples, 15 good apples

$$p = \frac{1}{6}, q = \frac{5}{6}$$

$$\text{Variance} = npq = 3 \times \frac{1}{6} \times \frac{5}{6} = \frac{5}{12}$$

Question: A box contains 10 red balls, 30 White, 15 orange balls, 10 or 20 blue balls, 2 balls drawn with replacement. Find the probability such that the first drawn ball is red and second drawn ball is white?

Answer: $\frac{4}{75}$

Solution:

10 R, 30 W, 15 O, 10 or 20 B

$$P(1^{\text{st}} \text{ Red } 2^{\text{nd}} \text{ W}) = \frac{10}{65} \times \frac{30}{65} = \frac{12}{169}$$

$$\text{or } = \frac{10}{75} \times \frac{30}{75} = \frac{12}{225} = \frac{4}{75}$$

Question: Using the word “DISTRIBUTION “. Find the no of ways of selection of 4 letters

Answer: 191.00

Solution:

D, 3I, 2T, R, B, U, O, N, S

4 different: ${}^9C_4 = 126$

2 alike, 2 different: ${}^2C_1 \times {}^8C_2 = 2 \times 28 = 56$

3 alike, 1 different = $1 \times {}^8C_1 = 8$

2 alike, 2 alike = 1

Total = 191

Question: $\sum_{r=0}^n \frac{{}^nC_r \cdot {}^nC_r}{r+1} = \alpha$, $\sum_{r=0}^n \frac{{}^nC_r \cdot {}^{n+1}C_r}{r+1} = \beta$. If $4\beta = 7\alpha$, then find n .

Options:

(a) 2

(b) 4

(c) 6

(d) 5

Answer: (c)

Solution:

$$\sum_{r=0}^n \frac{{}^nC_r \cdot {}^nC_r}{r+1} = \alpha$$

$$\sum_{r=0}^n \frac{{}^nC_r \cdot {}^{n+1}C_r}{r+1} = \beta$$

$$\alpha = \frac{1}{n+1} \sum {}^{n+1}C_{r+1} {}^nC_r = \sum \frac{{}^{2n+1}C_n}{n+1}$$

$$\beta = \frac{1}{n+1} \sum {}^{n+1}C_{r+1} {}^{n+1}C_r = \sum \frac{{}^{2n+2}C_n}{n+1}$$

$$\frac{7}{n+1} \left(\sum {}^{2n+2}C_n \right) = \frac{4 \left(\sum {}^{2n+2}C_n \right)}{n+1}$$

$${}^{2n+1}C_n = \frac{4 \times 3n + 2}{n} \left(\sum {}^{2n+2}C_{n-1} \right)$$

$$\frac{7}{8} \frac{n}{n+1} \times \frac{{}^{2n+1}C_n}{{}^{2n+1}C_{n-1}} = 1$$

$$\left(\frac{2n+1-(n)+1}{n} \right) = \frac{8(n+1)}{7n} \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$$

$$\frac{n+2}{r} = \frac{8(n+1)}{7r}$$

$$7n+14 = 8n+8$$

$$n = 6$$

